Scalability & Future Plan

Project Demonstration

Date	17 July 2026
Team ID	Not provided
Project Name	House Rent Management System (HouseHunt / RentEase)
Maximum Marks	1 Mark

Current Limitations

Limitation	Impact	Priority
Hard-coded API base URL	Deployment requires source edits	High
Local image storage	Does not scale across multiple server instances	High
JWT authentication without role authorization	Any valid token may reach role endpoints	High
Password reset without identity verification	Account takeover risk	High
Database queries fetch all records before filtering	Higher memory and latency as data grows	Medium
No automated test suite	Regression risk	High

Scalability Plan

Area	Upgrade
Configuration	Use environment-based API URLs and a checked-in .env.example.
API	Add role middleware, validation, pagination, indexes, rate limiting, and centralized errors.
Data	Filter in MongoDB queries and index email, ownerId, userID, and propertyId.
Files	Move images to object storage with unique generated names.
Deployment	Containerize services and use managed MongoDB, HTTPS, logs, and monitoring.

Future Roadmap

Phase	Enhancement	Expected Impact
Phase 2	Notifications, saved searches, and better validation	Improved usability
Phase 3	Payments, agreements, and identity verification	Safer transactions
Phase 4	Maps, geospatial search, recommendations, and analytics	Better discovery
Phase 5	Cloud storage, CI/CD, observability, and load scaling	Production readiness